ITA0512 - Computer Vision.

CO2 - AT3 - Class Test.

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23/25

11. An Image contains high frequency Noise.

- High-Frequency noise appears as rapid intensity variations that reduce image quality and visual clarity
- A Low-pass Filter (LPF) is used because it removes unwanted high-frequency components
- The image is transformed into the frequency domain using the Fourier Transform (FFT)
- The LPF suppresses high frequencies while preserving important low-frequency image information.
- The filtered image becomes smoother, less noisy, and more suitable for further image processing.

12. preserve low-frequency components while removing noise.

- Low-frequency components contain the important structure and smooth regions of an image.
- The Lowpass filter is applied to preserve these components while removing high-frequency noise.
- The image is first converted into the frequency domain using the Fourier Transform.
- High-frequency components are attenuated, while low frequency components are allowed to pass.
- The resulting image has reduced noise and retains its overall visual appearance and important details.

13. An Image requires edge enhancement in the frequency domain.

- Edge enhancement improves object boundaries by emphasizing sudden intensity changes in the image
- A High pass Filter (HPF) is used because edges contain high frequency information
- The image is transformed into the frequency domain before applying the filter.
- The HPF suppresses low-frequency components while preserving and enhancing high-frequency components.
- The output image contains sharper edges, making object detection and feature extraction more effective.

5

14. A smooth version of an Image is required.

- Image smoothing reduces noise and unwanted intensity variations present in the image.
- A low pass filter is selected because it smooths the image by removing high frequencies.
- The image is converted into the frequency domain using the fourier Transform.
- High-frequency components are reduced while low-frequency components are preserved during filtering.
- The final image appears smoother, less noisy, and better suited for image analysis

5

15. An object is clearly brighter than the Background.

- The Threshold segmentation is suitable because the object has higher intensity than the background.
- A Threshold value is selected to separate the object pixels from background pixels.
- pixels above the threshold are classified as object, while others become the background.
- This technique is simple, fast, and computationally efficient for image segmentation tasks.
- The segmented image clearly highlights the object, improving clear highlights improving object detection and image analysis.

5